

JUSTIN DESSUREAULT

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PROFILE

- Automation & Robotics engineering student with hands-on experience in embedded systems, EMC testing, product safety, and automation. Experienced with Python, ESP32, IoT integration, SolidWorks, and 3D printing. Passionate about combining electronics, programming, and mechanical design to develop innovative engineering solutions.

TECHNICAL SKILLS

- Programming & Software:** Python, C, C++, HTML/CSS, JavaScript, Node.js, Git/GitHub
- Engineering & Test Software:** PyVISA, SCPI Instrument Control, VS Code, SolidWorks (CAD), Cura, Microsoft Office, Technical Documentation
- EMC & Product Safety:** EMC Pre-Compliance Testing, Radiated Emissions Testing, Conducted Emissions Testing, ESD Testing, Spectrum Analyzers, Regulatory Test Procedures, Root Cause Analysis, Troubleshooting
- Hardware & Embedded Systems:** ESP32, Arduino, Sensors, Servos, Pumps, Motor Drivers, LCDs, Serial Communication (UART), USB Communication, IoT Systems, Embedded Programming, Robotics, Automation
- Design & Manufacturing:** 3D Printing, Mechanical Design, Prototyping

PERSONAL PROJECTS

- Automated Plant Watering System** – Built IoT device with ESP32, capacitive soil sensors, servo motor, water pump, and LCD. Developed custom local/remote web interface for monitoring and control.
- Robotic Arm** – Designed in SolidWorks with 3D-printed components; integrated multi-servo control, inverse kinematics, and a pan/tilt camera for 5×5 ArUco marker tracking.
- Portfolio Website** – Designed and deployed personal portfolio site using HTML, CSS, JavaScript, GitHub Pages, and Cloudflare tunnels. (stellaz.org)

EDUCATION

Bachelor of Engineering (Automation & Robotics) Algonquin College, Ottawa	2024 - Present
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Electrical Engineering Technician Diploma Algonquin College, Ottawa Dean's Honour List	2019 - 2022
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Social Science Diploma

2016 - 2019

Heritage College, Gatineau

- Business Administration

WORK EXPERIENCE

Product Safety & EMC Specialist Co-op

Jan 2026 – Apr 2026

Ross Video, Ottawa, Ontario

- Conducted EMC pre-compliance testing on broadcast video equipment, including radiated emissions and ESD testing
- Operated and configured laboratory equipment including spectrum analyzers and automated test systems
- Developed Python-based automation tools to improve radiated emissions testing efficiency using SCPI instrument control and turntable integration
- Assisted with product safety and EMC documentation, test procedures, and compliance-related reporting
- Collaborated with engineering teams to support third-party product certification and compliance requirements

Personal Care Aide/Health Care Aide

Jun 2023 – Dec 2024

Quebec Government, Gatineau

- Supervised medication taking

Customer Service Representative

Jan 2021 - Jun 2023

Couche-Tard, Aylmer

- Provided customer service, trained new staff, and handled cash and store operations

Planning and Performance Assistant

Jul 2018 - Dec 2018

Immigration, Refugees and Citizenship Canada, Ottawa

- Developed dashboards, reports, and data visualization products
- Analyzed data and statistics to support planning activities
- Created instructional guides for internal tools and processes